**NORTHSTAR ENVIRONMENTAL**
Water and environmental testing
Bloomington, Minnesota · (952) 555-0122 · northstarenviro.com

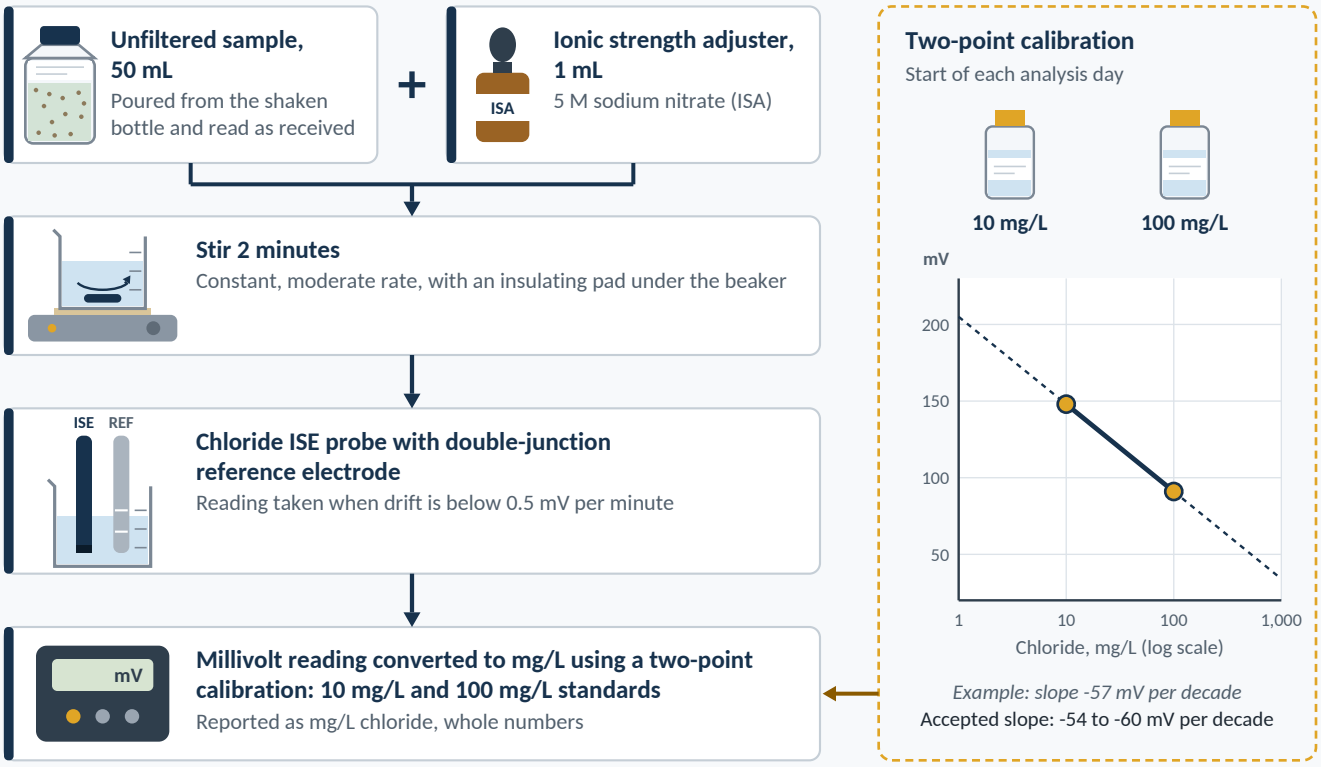
METHOD SHEET
**NE-CL-ISE-02**
Revised 2024-03-12

# Chloride by Ion-Selective Electrode

*Fast, affordable chloride testing with results comparable to EPA 300.0.**

Our chloride screen reads chloride directly in the water sample with a chloride ion-selective electrode (ISE). Samples are not filtered or run through a chromatograph, and most lake, stream and storm water samples need no dilution, so each one is read in a few minutes and the price stays low. Typical uses: road-salt and snowmelt screening, storm sewer inlets and outlets, private wells and process water.

**Figure 1. Sample workflow**




## METHOD AT A GLANCE

**Technique:** chloride ion-selective electrode, direct reading

**Electrode range:** about 2 to 35,000 mg/L (manufacturer specification)

**Sample volume:** 100 mL minimum, no preservative

**Holding time:** 28 days from collection

**Calibration:** two standards, 10 and 100 mg/L, run at the start of each analysis day. Acceptable slope -54 to -60 mV per decade.

**Quality control:** calibration check at the start of each day.

**Reporting limit:** 10 mg/L

**Turnaround:** about 10 days

* Screening method. Not accredited for chloride.

## 1. Scope

Chloride in lake, stream, storm and ground water and in process water. This is a screening method for monitoring and comparison work. It is not used for drinking-water compliance or permit reporting.

## 2. Principle

The solid-state sensing element of the chloride electrode develops a potential that follows the logarithm of chloride activity:

$$E = E° + S \log C$$

E is the measured potential in millivolts, C is chloride in mg/L and S is the electrode slope (theoretical value -59.2 mV per decade at 25 °C). Ionic strength adjuster (ISA) goes into every sample and standard in the same proportion, so the reading follows concentration. The meter calculates E° and S from the day's two standards and converts each reading to mg/L.

## 3. Sample collection and handling

- **Container:** 250 mL plastic bottle supplied by Northstar. Fill to the shoulder and cap tightly.
- **Volume:** 100 mL minimum.
- **Preservation:** none. Keep samples cool and bring them in on ice.
- **Holding time:** 28 days from collection.
- **Filtration:** none. Samples are analyzed as received; do not filter.

## 4. Procedure

**Equipment:** chloride ISE and double-junction reference electrode (outer fill 10% potassium nitrate), benchtop meter reading to 0.1 mV, magnetic stirrer with insulating pad, 100 mL beakers.

1. Let samples and standards reach room temperature, within 1 °C of each other.
2. Shake the sample bottle and pour 50 mL into a 100 mL beaker.
3. Add 1.0 mL of ISA (5 M sodium nitrate) and a stir bar.
4. Stir at a constant, moderate rate for 2 minutes, with the insulating pad between the stirrer and the beaker.
5. Rinse the electrodes with deionized water, blot dry and lower them into the sample.
6. Record the reading when drift is below 0.5 mV per minute. The meter displays chloride in mg/L.
7. Rinse and blot the electrodes before the next sample.

## 5. Calibration

- **Standards:** 10 mg/L and 100 mg/L chloride, made up from a 1,000 mg/L chloride stock standard.
- Each standard is prepared and read like a sample: 50 mL plus 1.0 mL of ISA, stirred 2 minutes.
- Calibrate at the start of each analysis day, reading the 10 mg/L standard first.
- **Acceptable slope:** -54 to -60 mV per decade. Outside this range, clean the electrode and repeat the calibration.

## 6. Quality control

| Check | Frequency | Acceptance |
|---|---|---|
| Two-point calibration | Start of each analysis day | Slope -54 to -60 mV per decade |
| Calibration check: 100 mg/L standard read as a sample | Start of each analysis day, after calibration | 90 to 110 mg/L |

Matrix spikes and sample duplicates are not part of this screening method.

## 7. Interferences and limitations

- Bromide, iodide, sulfide and cyanide respond at the electrode and read as chloride. Ammonia and thiosulfate also cause errors.
- A 1 °C difference between samples and standards causes an error of about 2%.
- Oils and suspended solids can coat the sensing element and slow its response. Rinse between samples and polish the electrode when the response slows.

## 8. Reporting

Results are reported in mg/L chloride, rounded to whole numbers. Each report lists the analysis date, the calibration standards and the electrode slope. Reporting limit: 10 mg/L.

## 9. Ordering and drop-off

Sample bottles and chain-of-custody forms are at the front desk; give your name. Bring samples on ice, Monday to Friday before noon, and sign the custody form at drop-off. Results in about 10 days.

**Questions:** Gary Lindqvist, Laboratory Director, (952) 555-0122, glindqvist@northstarenviro.com

**Approved:** G. Lindqvist, Laboratory Director

Revised 2024-03-12